

The question

Every contact number MarFold publishes comes from a 554-protein eval set whose FoldBench half is the **first 100 rows** of `monomer_protein.csv` — chosen before we understood how much of the eval set our training corpora contained (#213, #225). #232 then trained the #199 recipe from scratch on corpora decontaminated against **all** of FoldBench, which makes the other 234 monomers a real held-out test set for those checkpoints for the first time.

So: cut FoldBench's 334 monomers into three sets, score them, and ask whether the set we have been reporting on was telling us the truth.

- **eval-val (97)** — the natural monomers inside the historical FoldBench-100. - **eval-test (217)** — every other natural FoldBench monomer, never scored here. - **eval-denovo (19)** — every de novo designed FoldBench monomer.

Every protein carries a viral flag, because #241 showed the two strata rank models differently.

What was checked before anything was scored

****Decontamination, five links, all verified.**** All 334 monomer sequences are in #225's decontamination reference byte-for-byte; 131,180 training rows match one of them at the applied rule ($\geq 30\%$ identity over $\geq 50\%$ of the shorter sequence) and ****all 131,180 are in the drop list — zero survive****; the published corpora match #225's counts; #232's tokenizer pins those exact counts and those exact bucket prefixes; and both W&B runs read only those caches.

****What the rule does not cover, priced rather than described.**** At the applied gate the highest surviving identity to any of the 334 is 0.299. Relax the coverage requirement to 40 % and essentially every protein has a surviving training relative at $\geq 30\%$ identity; with no coverage requirement, 65 of them have one at $\geq 90\%$ over a fragment. "Decontaminated at 30 %" means that rule, not "no shared subsequence".

****Ground truth rebuilt through one path****, with the 126 overlapping frozen units reproduced byte-identically. ****One protein excluded****: `8uxt\_A` (1,596 residues) has no representable contacts-v1 document at an 8,192-token context.

The evaluation reproduces PR #244

Two of the three checkpoints are the ones #244 scored, and all 97 eval-val proteins are inside #244's universe, so the path is validated against a published reference protein by protein rather than against a tolerance on an aggregate: mean R-precision differs by ****−0.0031**** (m2-p06) and ****−0.0023**** (m1-p02) with per-protein $r = 0.996$ — inside the 0.0023 spread #204 measured for one unchanged checkpoint. 333 units × 3 checkpoints, ****0 unfinished rollouts****.

The baseline path has its own exact control: re-scoring 12 published proteins through this experiment's rebuilt ground truth reproduces #213's ESMFold and ESMFold2 numbers with ****max absolute difference 0.0****.

The result: eval-val was not flattering us

All-range R-precision, quoted as ****eval-val (97) / eval-test (217) / eval-denovo (19)****.

The two decontaminated #232 checkpoints: ****m2-p06 0.520 / 0.538 / 0.591**** and ****m1-p02 0.473 / 0.493 / 0.588****. The contaminated reference, #199's cooldown: ****0.589 / 0.613 / 0.619****.

Baselines on the same proteins: Protenix-v2 single-seq 0.263 / 0.265 / 0.835; ESMFold 0.750 / 0.753 / 0.795; ESMFold2 0.802 / 0.792 / 0.864; Protenix-v2 + MSA 0.846 / 0.845 / 0.844. The sequence-KNN null scores 0.584 / 0.582 / 0.066 over the unfiltered corpus and 0.407 / 0.426 / 0.050 over the decontaminated one.

****Every predictor scores the same or slightly better on the 217 monomers we had never touched****, and the contaminated reference moves the same way and by the same amount as the decontaminated checkpoints: the difference-in-differences is ****−0.006**** (m2-p06) and ****−0.004**** (m1-p02), an order of magnitude inside the noise. H2 is not supported — the historical FoldBench-100 was not inflating #199's score through memorised homologs. H1 and H3 hold: every predictor moves by under 0.03 between the two sets.

Two things this changes

****The KNN null is the yardstick, and it is corpus-specific.**** Copying the contacts of the ten nearest training sequences scores 0.582 on eval-test out of the unfiltered corpus and 0.426 out of the decontaminated one — 0.156 of memorisable contact map per protein, over 0.2 for 99 of the 314 natural monomers. Each model clears the null over the corpus it trained on (#232 m2-p06 ****+0.112****, #199 cooldown ****+0.031****) and falls below the null over the richer one. #199's 0.075 lead is therefore not a measurement of what decontamination costs — the runs are not budget-matched, 290,400 vs 145,199 steps — but it bounds how much of any contacts-v1 score is reachable by memorisation.

****Protenix-v2 single-sequence is not a comparator on natural proteins.**** 0.835 on the 19 designs, 0.265 on the 314 natural monomers. The "parity with Protenix-SS" framing came from an eval set that is three-quarters designed protein; on natural FoldBench monomers both #232 checkpoints beat it by more than 0.27. The real gap is to ESMFold2 (−0.255) and Protenix + MSA (−0.307).

****Viral proteins are harder for everything except MSA**** — eval-test viral vs non-viral: m2-p06 0.465/0.542, #199 cooldown 0.497/0.621, ESMFold2 0.608/0.804, seq-KNN 0.262/0.602, Protenix + MSA 0.812/0.847. #241's finding survives on new proteins, and the gap tracks reachable homology.

Designs are easier — against the right natural set

Earlier work (#213, #226, #241) found de novo designs much easier than natural proteins. Here the design advantage looks small: **$+0.054$** for #232 m2-p06 and **$+0.006$** for the #199 cooldown against all of eval-test. Both statements are true, about different comparisons.

The published contrast used exp65's 396 idealised `denovo\_pdb` designs against the **homology-filtered** natural set (eval2-natural, where MarFold scores $\sim 0.31-0.36$). exp245's eval-denovo is instead the 19 synthetic monomers FoldBench happens to contain — engineered binders and miniaturised folds, $n = 19$, interval ± 0.09 — and eval-test is *every* natural monomer, only 23 of 217 of which are under 40 % identity to #199's training sequences.

Split eval-test on that axis and the old pattern returns exactly: against natural proteins with no close training homolog, designs are **$+0.177$ [$+0.044$, $+0.306$]** easier for m2-p06 and identically $+0.177$ for the cooldown. Designs are much easier than natural proteins we have no homolog for; they are about as easy as natural proteins in general, because most natural proteins have homologs.

Protenix-v2 single-seq is the extreme: **$+0.570$** designs versus all natural, and flat across the identity split (0.243 vs 0.267). It looked competitive only on an eval set that was three-quarters designed protein.

Viral proteins: the penalty is ordered by homology dependence

Pooled over the 314 natural monomers (295 non-viral / 19 viral), the drop from non-viral to viral is: seq-KNN over the unfiltered corpus **$**-0.351**$** , seq-KNN over the decontaminated corpus -0.229, ESMFold2 -0.170, ESMFold -0.153, #199 cooldown -0.123, #232 m1-p02 -0.100, #232 m2-p06 -0.076, Protenix-v2 + MSA **$**-0.045**$** , Protenix-v2 single-seq **$**-0.002**$** .

That ordering is the finding: the penalty tracks how much a predictor leans on homology. A KNN lookup is nothing but homology and loses a third of its score. ESM's stems are trained on databases where viruses are thin. Protenix single-sequence has no homology signal to lose, and Protenix + MSA fetches one at inference time. An MSA is what closes the viral gap.

The caveat is size: FoldBench's monomers are 5.7 % viral, 19 in total and only 6 in eval, where intervals span about ± 0.15 . Only the pooled cell supports an argument. Growing this stratum means sampling recent PDB directly, as #241 recommended.

eval-denovo is 19 because that is all FoldBench has

FoldBench is 1,493 PDB entries across seven task files, but six of them are interfaces, and designed protein is rare throughout: **43** designed entries in total, 19 of them in the monomer task. The 24 outside it are mostly designed binders in complex, where a monomer contact eval would be scoring a chain solved with its partner present.

So eval-denovo is a sanity check, not a designed-protein benchmark -- and the obvious substitute does not work either. exp65's `denovo\_pdb` (396 designs inside the legacy 554, already fully scored) has **50.5 %** of its structures deposited on or before Protenix-v2's 2021-09-30 cutoff, and 43 % predate 2020-05, so they sit in the baselines' training data. A MarFold-versus-baseline number there is contaminated *for the baselines*.

The rule that follows: a set used against baselines has to postdate the baselines' cutoffs, not just our own corpora. Decontamination has two sides and we control one. FoldBench satisfies this by construction -- 0 of eval-test's 218 proteins predate Protenix-v2's cutoff -- and it retroactively qualifies #226's "designs are much easier" finding, which was measured on a set half of which Protenix-v2 trained on.

Use from here

The three sets have different **read budgets**, which is the point of the split.

`eval-val` (97) is the working set — checkpoint selection, sweeps, mid-training curves, anything routine. `eval-test` (217) is a held-out confirmation set, read rarely and logged in `data/eval\_test\_reads.md`; this experiment's construction read is entry 1 and nothing was selected on it. `eval-denovo` (19) keeps designs out of natural-protein means.

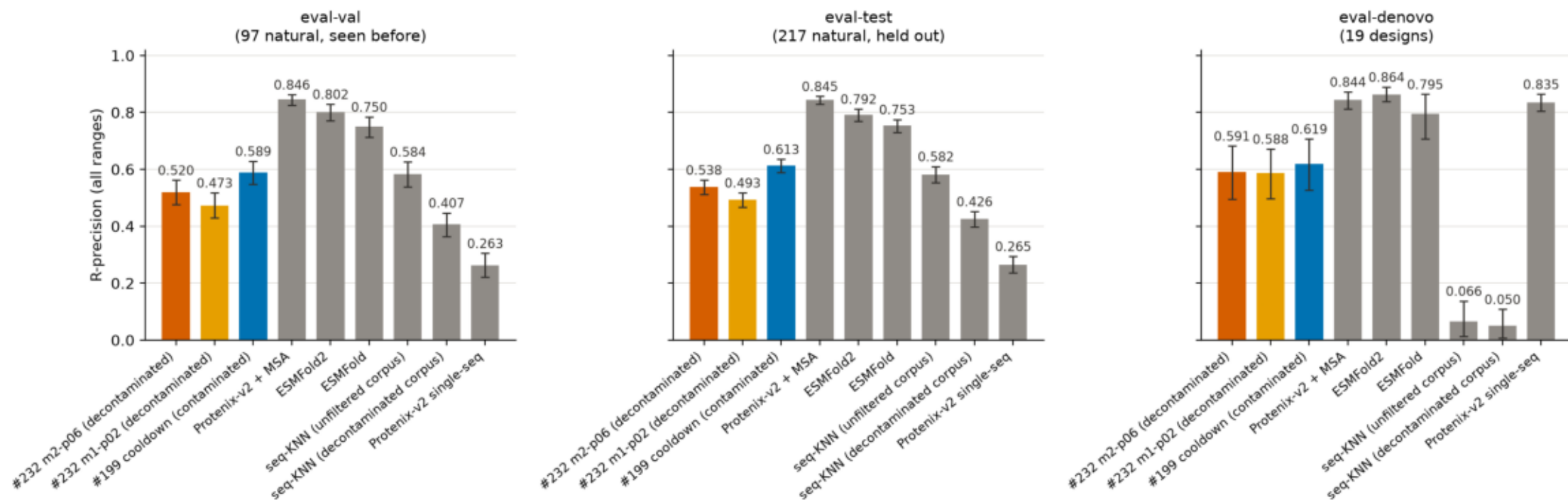
The result above is what makes that division workable: every predictor lands within 0.03 of the same number on the two natural sets, so eval-val is an unbiased stand-in for held-out performance and does not need checking against eval-test "just in case". If the read ledger grows a routine tail, eval-test is spent and needs replacing by sampling recent PDB directly (#241).

Everything is on the bucket under `data/contacts-v1-foldbench-monomers-exp245/`.

eval_sets_scoreboard.png

All-range R-precision per predictor on each of the three FoldBench monomer eval sets, with 95 % bootstrap intervals over proteins. MarFold checkpoints coloured (orange/amber = the decontaminated #232 pair, blue = the contaminated #199 cooldown reference), baselines grey.

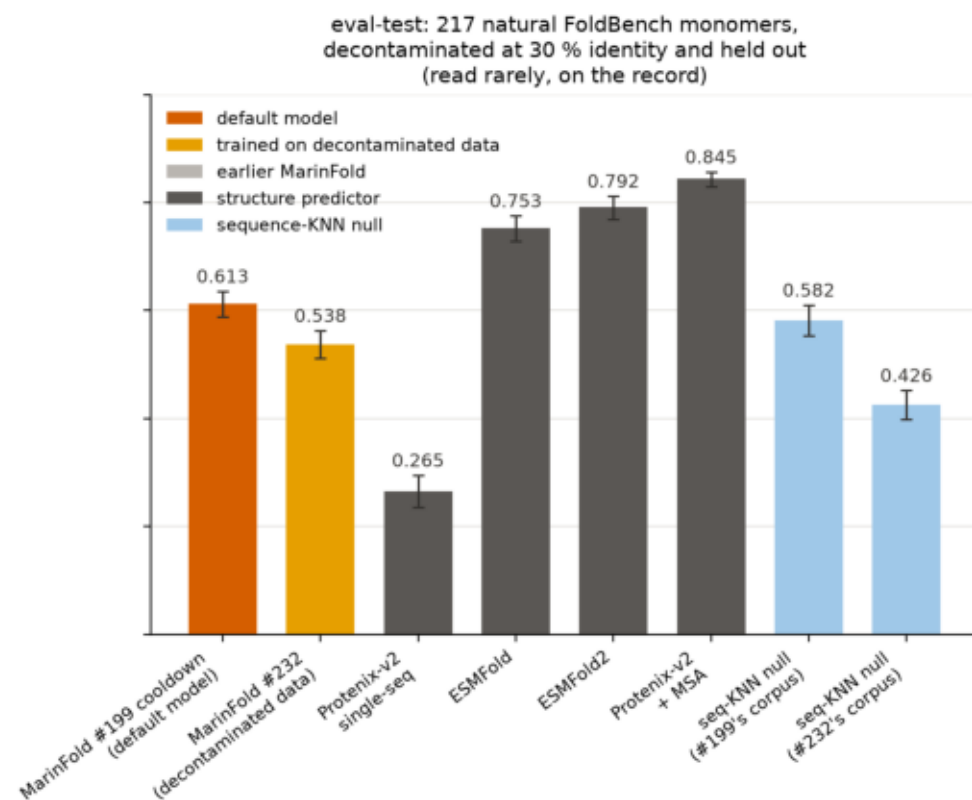
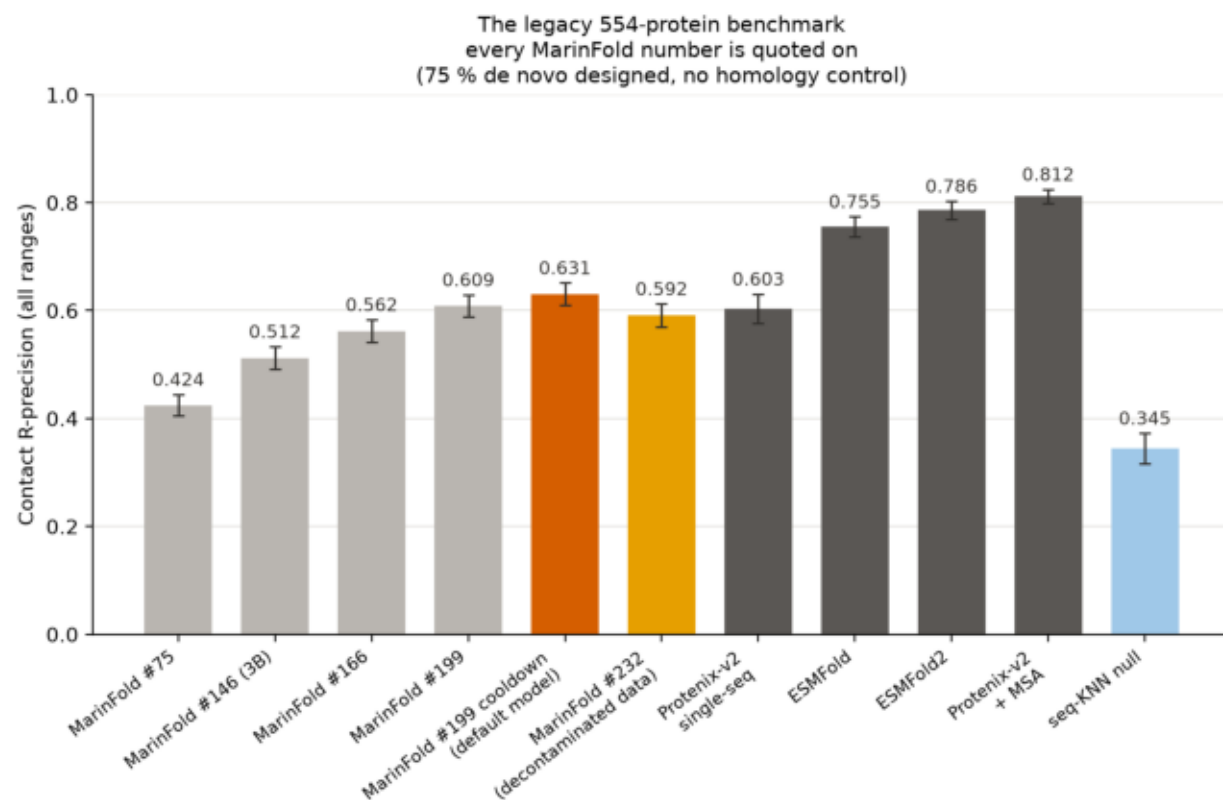
Contact R-precision on FoldBench's monomers, split into what we had already scored and what we had not



readme_performance.png

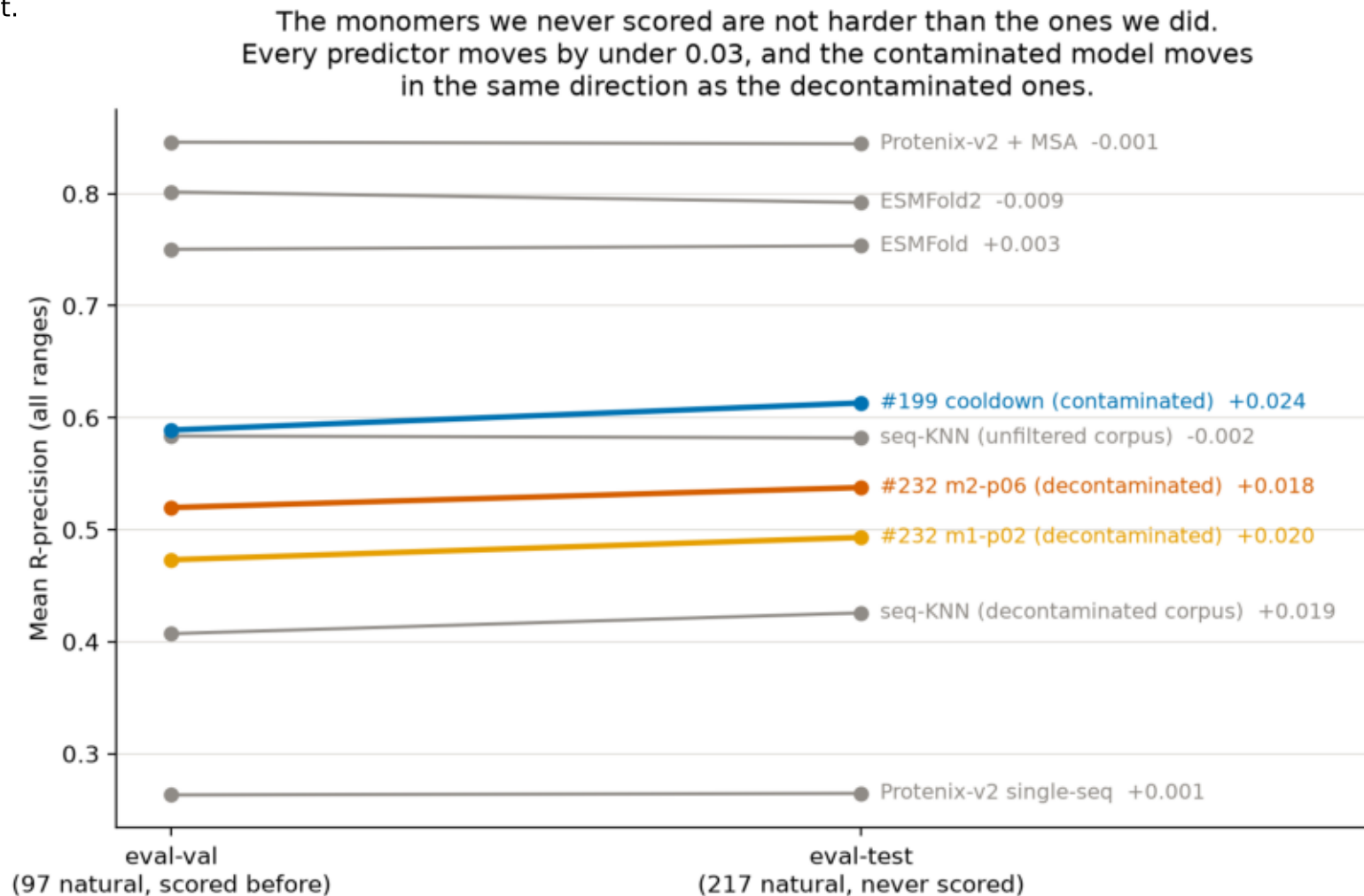
Contact R-precision on the legacy 554-protein benchmark (left, where every published MarFold number lives) and on exp245's eval-test (right, 217 natural FoldBench monomers held out and decontaminated at 30 % identity). Error bars are 95 % bootstrap intervals over proteins. The two sequence-KNN nulls copy the contacts of each protein's ten nearest training sequences, out of the corpus each model family actually trained on.

Predicting residue-residue contacts from a single sequence, no MSA and no PLM



val_vs_test.png

Each predictor's mean R-precision on eval-val (the 97 natural monomers we have always scored) joined to eval-test (the 217 we never had). Every slope is under 0.03 and the contaminated model's is not steeper than the decontaminated ones', which is the experiment's main result.



viral_split_scoreboard.png

All-range R-precision per predictor, split viral vs non-viral, on each natural eval set and on the two pooled. Hatched bars are viral. Error bars are 95 % bootstrap intervals; the viral cells hold 6, 13 and 19 proteins, so their intervals are wide by construction. eval-denovo is omitted -- none of the 19 designs is viral.

Viral proteins are harder for every predictor that depends on homology — and not for Protenix + MSA

